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Research papers

ERT and salinity – A method to determine whether ERT-detected preferential pathways in brackish water-irrigated soils are water-induced or an artifact of salinity

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ABSTRACT

The identification of preferential flow generation and the tracking of their pathways in agricultural soils are challenging tasks, in particular while keeping the soil profile undisturbed. The noninvasive electrical resistivity tomography (ERT) method seems, a priori, to be an ideal tool for these purposes, but the simultaneous dependence of soil electrical resistivity (ER) on soil moisture content and salt concentration poses a problem when high-salinity water is involved. The dependence of these variables is expressed, for example, by Archie's law. Irrigation with brackish or treated wastewater are two such complicated cases. The latter has been found to render soils water-repellent with inherent flow in preferential pathways. A method that resolves this complication is developed and applied for in-situ-measured ERT data. This method combines a model that simulates the preferential ER(t) as a series of interconnected well-mixed units (WMUs) with frequent ERT scans during soil wetting and subsequent drying. Following validation by comparison of its output to simulations using the Richards equation for flow and convection-dispersion equation for transport, the WMU model was used to simulate scenarios of simultaneous variations in moisture content, salinity, and ER(t) in a well-mixed soil volume (unit). The characteristic patterns acquired by these simulations were compared with in-situ measured ER(z,t) to determine whether the observed preferential ER pathways coincide with preferential flow or salinity pathways, or both. From the eight sections in the profile where this analysis was made for, five exhibit preferential flow, characterized by fast wetting front propagation that includes by-passing of a certain soil volume, while the wetting front velocity in the other three was moderate. We, therefore, concluded, based on the water content/salinity effect on ER, that the measured preferential ER pathways coincide with preferential flow pathways. Salinity has, for the current case, a minor, if any, effect on the ER spatial variation. Therefore, ERT is an advantageous mean of mapping of preferential flow pathways in the soil profile, and the method developed herein can be used for studying the water distribution in the soil profile. Particular irrigation design and management can be then used to ease the negative effects of preferential flow pathways on soil water availability to plant roots and the enhanced leaching of agrochemical below the active root zone.

1. Introduction

Preferential flow pathways are one form of uneven water flow in the root zone. This flow pattern can form under a variety of circumstances, e.g., the presence of differently sized pores and macropores (structural cracks, rootholes, wormholes, etc.), or coarse-textured soils (Jarvis et al., 2016; Ritsema et al., 1993). Preferential flow can significantly affect infiltration-related processes, such as groundwater recharge, crop irrigation, vadose zone transport of nutrients and contaminants, and residence time of water in the vadose zone, where chemical compounds are degraded or exchanged in the soil. Recent studies have found preferential flow pathways (fingering) where the soil is water-repellent or even subcritically water-repellent (Bauters et al., 1998; Dekker and Ritsema, 1996, 2000; Ritsema and Dekker, 1994; Robinson, 1999; Täumer et al., 2005, Wallach et al., 2013). The different factors that render soil water-repellent are described in Doerr et al. (2000) and references therein. Moreover, Wallach et al. (2005) recently found that long-term irrigation with treated wastewater (TWW) renders the soil water-repellent, inducing the formation of preferential flow pathways.

Crop irrigation with TWW has become a common practice in arid and semiarid areas to deal with water scarcity and to reduce the usage of fresh water. In Israel, the reuse of former wastewater already

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provides more than 50% of total water consumption by agriculture (OECD, 2015). Besides the advantages as a sustainable water resource and recycling of nutrients, compared to fresh water, TWW is generally characterized by a higher load of dissolved organic matter, suspended solids, sodium adsorption ratio, and significant levels of salinity. Compounds that can render a surface hydrophobic and that have been identified in water-repellent soils are broadly subdivided into aliphatic hydrocarbons and amphiphilic compounds, mainly long-chained fatty acids (Franco et al., 2000; Horne and McIntosh, 2000). The latter compounds were also suggested to play a role in the development of water-repellent in soils irrigated with wastewater (Graber et al., 2009). A reduced affinity of soil to water, so-called soil water repellency, may lead to reduced infiltration capacities, intensify overland flow, and formation of preferential flow paths that render the spatial wetting pattern uneven (Wallach et al., 2005; Rahav et al., 2017).

Under these conditions the preferential flow is considered as gravity induced unstable flow, resulting in vertical narrow paths that have been noted as fingered flow (Glass et al., 1989). However, identifying the extent of preferential flow pathways in the soil profile is a challenge. The difficulty is related to the methods used to measure uneven soil water distribution in the root zone and its variation with time, based on installing many sensors at predetermined or random locations in the soil profile. The typically limited sampling volume of these sensors (10–100 cm^3) (e.g., Ferre et al., 1998) limits the estimation of spatial and temporal distribution of water content in the root zone (Bogena et al., 2007; Flury et al., 1994; Kim et al., 2008). Moreover, finding a priori unknown locations of preferential flow pathways using these sensors is not feasible (Scanlon, 2000; Vereecken et al., 2008). Furthermore, installation of the local-measurement sensors requires drilling (Gee et al., 2003), which interferes with the natural soil structure and water flow regime (Rothe et al., 1997). The advantages of non-invasive methods to measure soil water content are obvious.

Recently, electrical resistivity tomography (ERT) has been used in soil science studies as a noninvasive method for acquiring spatio-temporal distribution of soil profiles. It has been found to be an appropriate alternative to the aforementioned methods (Michot et al., 2003; Moreno et al., 2015; Nijland et al., 2010). In ERT, electrodes are inserted into the soil surface. A pseudo-direct electrical current (DC) is injected through two electrodes (current electrodes), and the resultant electrical potential is measured by another two electrodes (potential electrodes). The current flow in the subsurface is described by the Poisson equation, derived from a combination of Ohm's and Kirchhoff's laws (Loke, 2000). For 2D ERT surveys, the electrodes can be placed in a line at the soil surface, enabling a large number of possibilities for current and potential electrodes. First, a pseudo-section of the soil's electrical properties is constructed, and for each electrode quartet, a measurement's horizontal and vertical location is determined. The pseudo-section usually does not show the correct electrical resistivity (ER) distribution because each resistivity measurement is influenced by a larger soil volume than that around each measurement location, and the heterogeneity of the soil electrical properties with its effect on the electrical current in the soil (Furman et al., 2002). The soil heterogeneity causes this apparent resistivity to differ from the true one. To determine the true bulk ER of the soil, an inversion of the measured values must be conducted. In this inverse problem, the main goal is to determine the subsurface structure (i.e., the bulk ER distribution) that will provide a forward simulation result similar to the measured data, i.e., finding an ER distribution model that best explains the results (Loke, 2000). This is by definition an ill-posed problem. Therefore, solving it entails the application of some constraints, typically requiring smoothness of the solution (Loke, 2000).

Each ERT survey provides detailed distribution of the bulk soil ER over a relatively large soil volume (Binley et al., 1996; Michot et al., 2003). The strong dependence of bulk soil ER on water content makes ERT a relatively simple method to determine the spatial and temporal distribution of moisture content in the soil profile without disturbing it (Samouëlian et al., 2005). The semi-empirical Archie law (Archie, 1942) and its many variants (e.g., Waxman and Smits, 1968; Rhoades et al., 1976; Mualem and Friedman, 1991) have been used to relate bulk soil ER and soil water content. It is accepted that, in general, of the two, the ER is much more sensitive to the moisture content than to the salinity (Sheets and Hendrickx, 1995). Many studies on soil water content evaluation by ERT surveys have assumed a uniform spatio-temporal value of the pore water electrical conductivity (Beff et al., 2013; Srayeddin and Doussan, 2009; Zhou et al., 2001). However, the bulk soil ER's dependence on pore water electrical conductivity is disregarded, which may introduce errors when soil salinity is high, as in cases where brackish water or treated wastewater (TWW) are used for irrigation.

Recent studies on drip irrigation in a grapefruit orchard using ERT (Furman et al., 2013; Zur, 2009) have shown that an assumption of uniform pore water electrical conductivity leads to a larger apparent porosity moisture content by ignoring the salinity buildup as the irrigation season progresses. Moreno et al. (2015) isolated the contribution of pore water electrical conductivity from the bulk electrical conductivity by calibrating a coupled flow and transport model. The temporal dynamics of the measured electrical signal enabled a better understanding of the water and solute dynamics in the root zone. However, this method seems less practical when preferential flow paths are formed in a water-repellent soil profile by the flow instability associated with non-zero contact angles (Aminzadeh et al., 2011; Brindt and Wallach, 2017). The high water salinity associated with brackish water may then affect the interpretation of the measured bulk soil ER distribution. Thus, a priori determination of the coefficients in the modified Archie law that combines the water content and salinity effects on the bulk soil ER at the field scale is complicated and less unrealistic in practice.

Modeling of flow in gravity-induced fingers is still challenging the soil physics community (DiCarlo, 2013), and particularly fingers that are inherently formed in water repellent soils. Many efforts to simulate the formation of preferential flow paths and their extent using the 1D Richards equation have been unsuccessful owing to the lack of this parabolic type partial differential equation to simulate a sharp wetting front and the nonmonotonic soil water distribution that have been observed along gravity-induced fingers. More details on this issue can be found in DiCarlo (2013), and Brindt and Wallach (2017) and references therein. Therefore, the use the Richards equation-based model for a 2D preferential flow is not feasible, and alternative models should be developed and used. Moreover, the lack of a simple model, and given that soil resistivity is a function of both soil water content and salinity, an answer to the question whether a measured preferential ER path indicates a preferential flow pathway or spatial salt distribution of such shape by a 2D model is still open. Given that, this study provides a means to resolve this difficulty by combining measured ERT scans and a simplified 1D model for each ER preferential path.

Based on the above, the method suggested herein intends to examine whether a preferential ER path is formed mainly by a preferential flow pathway or rather by a spatial salt distribution. This method is based on already measured sequential preferential ER(t) paths and using a model that simulates the preferential ER(t) path as a series of vertical interconnected well-mixed units (WMUs). Frequent ERT in-situ scans during soil wetting and subsequent drying were used together with WMU simulations for ER variation during a complete irrigation cycle (soil wetting and drying) for different initial moisture content and salt concentration scenarios. The following stage includes a comparison of the ERT sequential scans with the WMU simulations that enables to deduce whether the ER preferential pathway results by a preferential flow, a similar shape of a spatial salt distribution or both. The method is used for an interpretation of actual ER data measured in water-repellent soils that are irrigated with brackish water. This study is complementary to a comprehensive field study by Rahav et al. (2017) that aimed to examine the effects of (i) soil-wettability decrease due to prolonged TWW irrigation on the spatial and temporal distribution of water content and associated chemical properties in the root zone; (ii) soil reclamation by replacing TWW irrigation with freshwater (FW) for soil reclamation.

2. Material and methods

2.1. Field study setup

The study took place in a commercial TWW-irrigated citrus orchard located in Sitriya, in central Israel. The orchard was planted in a non-calcareous, sandy loam soil (80.1 sand, 8.8 silt, and 10.1 clay), with an average pH of 7.2 (Rahav et al., 2017). The orchard was irrigated with a drip irrigation system with FW until 2008 when TWW from the Ayalon sewage treatment plant replaced the FW. The "Star Ruby" grapefruit trees were planted at spacings 2 × 6 m. In 2012, when the comprehensive study was initiated, an orchard area of 1500 m^2 was divided into two halves. One half continued to be irrigated with TWW, while FW replaced the TWW in the other half (Rahav et al., 2017). ERT surveys were performed to look into the spatial soil water distribution in the TWW and FW plots, with the expectation that these surveys would enable locating the preferential flow pathways and following their development. The plots and entire orchard were irrigated daily, excluding Saturdays, by a single drip line located 30 cm from the tree row. The 3.8 L/h drippers were 75 cm apart along the dripper lines, and the wetted area of adjacent drippers was expected to merge to form a continuous wetting strip along the dripper line. The average irrigation amount during the irrigation season (April to November) was 700 mm. The electrical conductivity of the FW was half that of the TWW (0.9 and 1.8 dS/m, respectively), both sources categorized as brackish water. For more details on the study site and its soil characteristics we refer the reader to Rahav et al. (2017)

2.2. ERT surveys

Successive 2D (x–z) ERT surveys, to trace the course of the temporal changes in ER distribution in the soil profile during wetting and subsequent drying, were performed for the TWW and FW plots. The SYSCAL Pro Switch 96 system (Iris Instruments, France) with 36 stainless-steel electrodes was used for these surveys. The electrodes were placed 15 cm apart in a line parallel to the tree rows and drip line (10 cm off the drip line). The surveys were started on the morning prior to irrigation initiation and stopped in the late afternoon hours, to monitor the spatial soil water distribution during the irrigation and subsequent drainage events. Each ERT survey included hundreds of separate electrical potential measurements using dipole-dipole (dp–dp) and Wenner–Schlumberger sequences. In the dp–dp sequence, single-electrode spacing was maintained between the current electrode pair as well as between the potential electrode pair. In the Wenner–Schlumberger sequences, a specific interval between the four current and potential electrodes was chosen for each measurement. Each measurement included 3 to 10 repeats of current injection and potential measurement. The preferred standard deviation of these repetitions was 1%, namely, if the standard deviation was higher after three repetitions, additional repetitions were conducted until the preferred standard deviation was reached or 10 reparation repetitions were completed. Measurements with larger than 10% standard deviation, even after ten repetitions, were ignored and removed from further analysis.

An inversion process is needed for achieving the ER distribution from the apparent resistivity measurements. This inversion aim is to provide a 2D ER distribution model, with a certain predetermined degree of uncertainty, based on the measured ER data. Owing to the solution nonuniqueness, the inversion process may lead to nonrealistic results (Binley and Kemna, 2005). To overcome this obstacle, additional constraints were added, and the inversion is treated as a regularized optimization problem (Tikhonov and Arsenin, 1977). A detailed explanation of the inversion process can be found in Binley and Kemna (2005). Data inversion for the apparent resistivity field measurements was performed with R2 software (Binley, 2013). This software provides an inverse solution to the measured apparent resistivity data based on a 3D charge current with a 2D finite element mesh. The solution is based on a regularized objective function combined with weighted least squares. The iterative optimization reduces the differences between measured resistivity values and those calculated with the inversion model until the difference, estimated by a root mean square error (RMSE) is reaching 1%.

2.3. Preferential ER pathway analysis

To determine whether a preferential ER pathway coincides with preferential flow pathways or, rather, represents a combination of salinity and water content distributions, the following steps were taken: (i) virtual 10-cm wide sections were plotted along what seemed to be preferential ER pathways; (ii) the sections were divided into three sequential segments of similar length (12 cm each); (iii) the ER variation with time for section i, $\mathrm{ER}_{i,j}(t)$, was determined from the successive ER surveys. Note that $j = 1, 2, 3$ denotes the segments in each column; (iv) the measured $\mathrm{ER}_{i,j}(t)$ and the well-mixed unit (WMU) simulations for the different sections (see Section 2.4 for details) were compared, to determine whether the preferential $\mathrm{ER}_i(t)$ coincides with a preferential flow pathway.

2.4. The WMU model

Following the WMU approach, a measured ER preferential pathway divided into a finite number of units within which the moisture content $\theta(t)$ and salt concentration $c(t)$ are uniformly distributed (Fig. 1). The WMUs are interconnected such that the water and salinity outflow from a unit is the inflow from unit underneath it. The upper unit represents the soil surface layer, and its input represents the boundary condition at the soil surface.

The WMU mathematical model is based on a mass balance of water and salt in each unit along the column. The model consists of the following stages (Fig. 1): (i) TWW or brackish water enters the upper unit at a rate Q_{1in} and salt concentration, c_{1in}. The initial water content in the upper unit, and in practice, between field capacity (FC), the water content in which water drainage turns negligible, and wilting point, should provide plants with available water to keep high yield. It depends on the farmer's experience, and soil and crop types. As long as $\theta_1(t) \le FC$, water does not leave the upper unit (actually any unit in the column) and is depleted by root uptake (transpiration), T, and evaporation. These two water depletion processes can be lumped in the model to a single evapotranspiration parameter, ET. The ongoing water and salinity inflow into the upper unit (given that $Q_{1in} > ET$) increases its water content and changes its salinity. (ii) Drainage, $Q_{1out} = Q_{2in}$, initiates when $\theta_1(t) >$ FC. The dissolved salt concentration in the water that leaves the upper unit is $c_1(t) = c_2(t)_{in}$. The initial water level in the lower unit can be either lower or higher than FC. For $\theta_2(t) <$ FC, water is depleted solely by root uptake, and for $\theta_2(t) >$ FC, water and salt leave the unit. Water content in this unit is then depleted by drainage and root water uptake simultaneously (Fig. 1). (iii) When irrigation stops, water and salt continue to drain as long as $\theta_i(t) >$ FC, where $i = 1$ for the upper unit and 2, 3 for deeper units. Note that water depletion by root water uptake continues. For $\theta_1(t)$, $\theta_2(t)$and$\theta_3(t) <$ FC, water and salt do not discharge out of the units while water depletion continues by root uptake. As the model simulations are run for an orchard that is irrigated daily, the water depletion by either ET or E is assumed constant as water content variation and free drainage, in particular, are less sensitive to hourly ET variation (Farahani et al., 2007; Wegehenkel and Beyrich, 2014 and references therein).. The balance equations for water and salt for coupled above and underneath

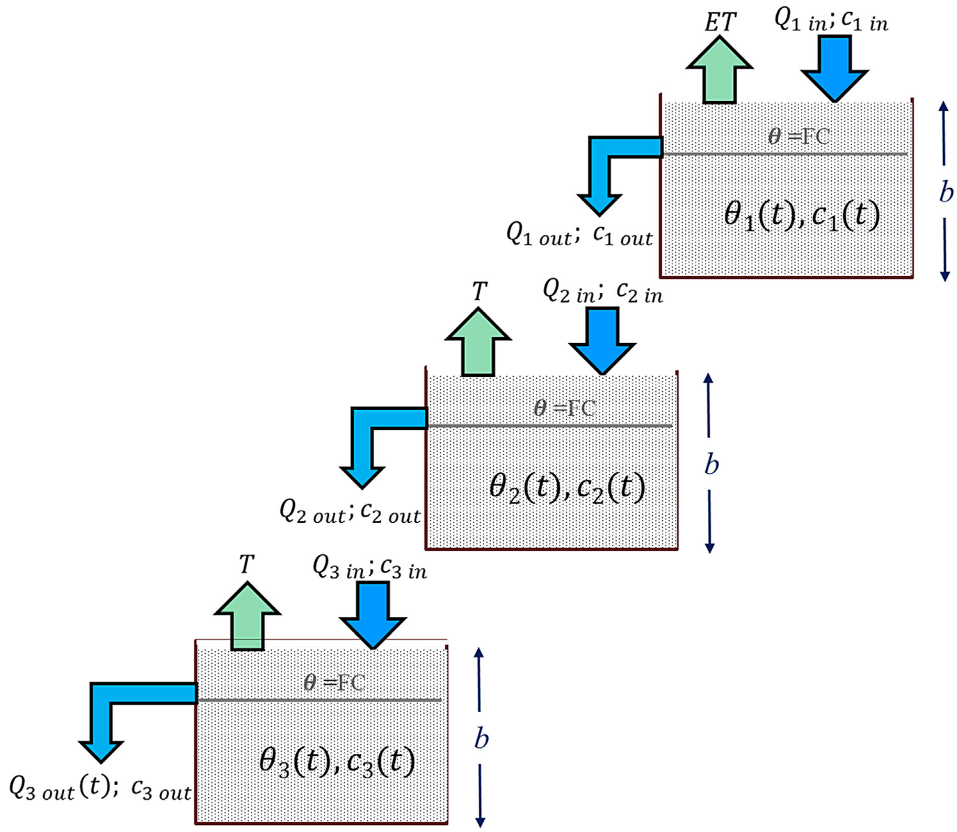

Fig. 1. Scheme of three well-mixed units (WMUs) representing three consecutive soil layers. The output of one unit is the input to the unit below it.

WMUs are:

For the upper unit, i = 1:

$$\begin{cases} b\frac{d\theta_1}{dt} = Q_{1in} - ET - Q_{1out}(t) \\ b\frac{d(\theta_{1i}\bullet c_{1i})}{dt} = c_{1in}\bullet Q_{1in} - c_1(t)\bullet Q_{1out}(t) \end{cases}; c_{1o} > c_{IR} \tag{1}$$

where b is the unit depth, Q_{1in} and c_{1in} are the water and salinity influx, respectively, and $Q_{1out}(t)$ is the outflow from the WMU, which depends on the difference between $\theta_1(t)$ and FC:

$$Q_{1out}(t) = \begin{cases} 0;\ \theta_{i0} \le FC \\ \alpha(\theta_1(t) - FC);\ \theta_{i0} > FC \end{cases} \tag{2}$$

in which α is the outflow rate coefficient, which indicates the relative water access (above FC) that will be drained in a single time step. Coarser soil and soil with a large number of macropores will have higher drainage parameters. The drainage expression as a whole when the moisture content is larger than FC in Eq. (2) can be considered as an analogous to the Darcy law under gravitation flow (unit gradient). For lower units i = 2, 3, ⋯ the influx is actually the outflux from the above unit. So replacing the influx, $q_{iin}(t)$, for lower units to the outflux from the above unit, $q_{i-1out}(t)$, yields:

$$\begin{cases} b\frac{d\theta_i}{dt} = Q_{i-1out}(t) - ET - Q_{iout}(t) \\ b\frac{d(\theta_i\bullet c_i)}{dt} = c_{i-1}(t)\bullet Q_{i-1out}(t) - c_i(t)\bullet Q_{iout}(t) \end{cases}; c_{i0} > c_{IR} \tag{3}$$

where again

$$q_{iout}(t) = \begin{cases} 0;\ \theta_{i0} \le FC \\ \alpha(\theta_i(t) - FC);\ \theta_{i0} > FC \end{cases} \tag{4}$$

These ordinary differential equations (Eqs. (1)–(4)) were solved in MATLAB 2016b (MathWorks, Inc., Natick, MA) using the built-in ode45 solver. The process was divided into a series of problems according to the different in/out fluxes at the units. Once a condition that triggers flux is reached (e.g., moisture content that reaches FC or irrigation initiation), a new problem is solved whose initial conditions are the moisture content and salt concentration at the end of the previous problem.

The Archie equation (Archie, 1942) was used to relate the simultaneous variation of $\theta(t)$ and $c(t)$ to $ER(t)$:

$$\rho(t) = \frac{kF}{c(t)}\left[\frac{\theta(t) - \theta_r}{\theta_S - \theta_r}\right]^{-m} \tag{5}$$

where$\rho(t)$ is the bulk ER, F and m are the formation and cementation factors, respectively, θ_S is the saturation moisture content and θ_r is the residual moisture content. The simulations in the sequel are made with F= 2.5 and m=1.3. The parameter k is a conversion parameter from solute concentration in meq/L to electrical conductivity in mS/m and was chosen to be equal to 100meqL^{-1}/Sm^{-1} (Rhoades et al., 1992).

3. Results

3.1. Sequential ERT surveys

The sequential ERT surveys that were measured in situ during TWW irrigation (soil wetting) and subsequent drainage (drying) are depicted in Fig. 2, where the ER distribution in the soil profile is seen to be spatially uneven. The relatively sharp changes in ER over short distances, particularly along the horizontal direction, are attributed to preferential pathways that were generated in the soil that was rendered water-repellent by the irrigation with TWW (Rahav et al., 2017). The high soil salinity in the plot (Rahav et al., 2017) may contribute to the spatial distribution of the ER(t) in general, and in particular to the formation of preferential ER pathways (Fig. 2). The difficulty in distinguishing between the relative contribution of high water content and salinity, as both simultaneously reduce the soil ER, led to the development of an indirect method based on a comparison between in-situ

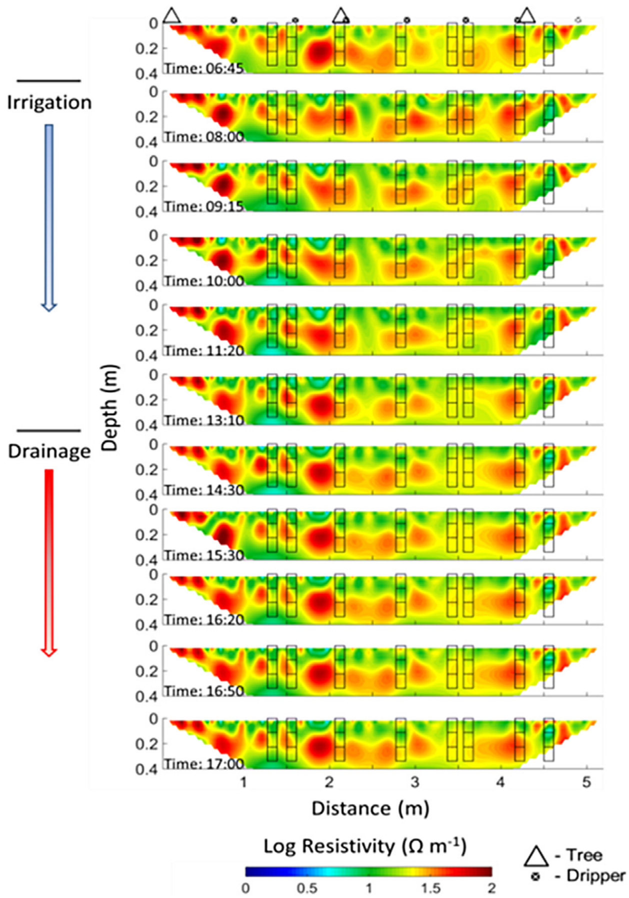

Fig. 2. Spatial ER distribution in the shallow root zone obtained by sequential ERT surveys performed before, during and following irrigation (drainage) in a TWW plot. The measured ER(t) was determined within the eight sections, numbered 1–8 from right to left in Fig. 6, located along the preferential ER pathways and compared to the WMU model simulations.

measured and simulated ER(t), using the WMU model for the latter.

3.2. WMU model verification

The WMU model was verified by comparing its simulation outcomes with simulations using the Richards (Richards, 1942) and convection-dispersion equations. The $\theta(t)$ and $c(t)$ in a single unit (Fig. 3) were calculated by averaging the data obtained by HYDRUS 1-D over the unit depth, b. The parameters used for the simulations (Eqs. (2) and (4)) were: $\theta_s = 0.38$; $\theta_r = 0.05$, van Genuchten parameters $\alpha_{VG} = 0.035$ and $n_{VG} = 1.94$, saturation hydraulic conductivity $K_s = 0.104$ cm/min, and the dispersion coefficient $D = 0.01$ cm^2/min for the HYDRUS 1-D simulations and: $b = 20$cm and $\alpha = 24 \times 10^{-4}$ min^{-1}, and FC = 0.16 for the WMU model. The soil hydraulic parameters for the HYDRUS 1-D simulations where taken from the Rosetta Neural Network Prediction algorithm provided by the HYDRUS 1-D software, based on the field study site particle size distribution (Rahav et al., 2017). The WMU model parameters were evaluated from the HYDRUS 1-D simulations. The FC was chosen when water depletion turned constant during drainage simulations. Namely, the drainage stops and water is depleted solely by ET. The alpha was then chosen to provide the best fit between the HYDRUS 1-D and WMU model simulations. The parameters for the WMU model have provided a good agreement between the WMU model and HYDRUS 1-D simulations. The simulations included two consecutive stages: (i) in Fig. 3a and b, water drained from an initially wet unit with brackish water ($\theta_0 = 0.34$ and $c_0 = 34$ meq/L) with a constant evapotranspiration (ET) of 5×10^{-4} cm/min. The drainage stage lasted 2880 min (2 days) for both models. In the WMU, the drainage stopped when water reached FC ($\theta = 0.16$), with only water depletion by ET occurring until $\theta \approx 0.12$. Very good agreement (RMSE = 0.007) was reached between the moisture content of the HYDRUS and WMU model simulations at the end of this stage (Fig. 3a). However, a slight deviation in salt concentration was obtained between the two models (Fig. 3b). (ii) At this stage, a flux of brackish water $q_{in} = 0.01$ cm/min and $c_{in} = 35$ meq/L was introduced into the unit for 3 h. The water content increased while the salt concentration decreased (Fig. 3c, d, respectively). Note that drainage in the WMU does not start before FC ($\theta = 0.16$) has been reached. The initial conditions for stage 2 were the final moisture content and salt concentration at the end of the former stage. As for the drainage scenario, a good agreement between the HYDRUS and the WMU model output was obtained with RMSE of 0.002 in the water content comparison (Fig. 3c), and 0.08 for the salt concentration (Fig. 3d).

The very good agreement between the WMU model and HYDRUS simulations (Fig. 3) enables a further use of the former to examine whether the measured preferential ER (Fig. 2) is generated by preferential water flow or a local salinity effect, and the extent of these effects.

3.3. Simulations for $\theta(t)$, $c(t)$ and $\rho(t)$ in a two-WMU system

Two case studies were simulated using the WMU model for two interconnected units (upper and lower) to study the typical variations in $\theta(t)$, $c(t)$ and $\rho(t)$ in a soil profile under brackish water irrigation. The WMU parameters and initial conditions for each case are provided in Table 1. The initial water content and salt concentration in the units (denoted by θ_{10} and c_{10} for the upper unit and by θ_{20} and c_{20} for the lower one) were similar for the first set of case studies. The second set included a scenario in which the initial water content differed among the two units ($\theta_{10} \neq \theta_{20}$). For both sets, with units lengths b was 20 cm, brackish water ($c_{in} = 9$ meq/L) entered the upper unit at a rate of $Q_{1in} = 0.03$ cm/min for 562 min. Water and salt outflow from the upper and lower units occurred as long as $\theta_1(t)$ and $\theta_2(t) > \text{FC} = 0.16$. Moreover, water depletion by plant root uptake occurred continually at a constant rate of 0.0055 cm/min (daily ET of 8cm/day). The coefficient for water outflow from the units was $\alpha = 0.01$ cm^{-1}, and the ER variation, $\rho(t)$, was calculated by substituting $\theta(t)$ and $c(t)$ into Archie's model (Eq. (5)).

Fig. 4 depicts the simulation results for the first set of case studies. The initial water content in both units was $\theta_{10} = \theta_{02} = 0.097$ ($<$ FC), whereas the salt concentration in the first case study was initially high ($c_{10} = c_{20} = 75$ meq/L) (Fig. 4b) and low for the second case study, $c_{10} = c_{20} = 25$ meq/L (Fig. 4e). The initial ER, ρ(t), in the units differed for the two case studies (Fig. 4c, f). Outflow from the upper unit started after 51 min when water content reached FC (Fig. 4a). Nevertheless, water content continued to increase as long as irrigation occurred. The water content in the lower unit initially decreased due to root uptake, and started to increase afterward when water inflow (outflow from the upper unit) exceeded the root uptake rate (Fig. 4d). As irrigation water salinity (0.09 meq/L) was lower than the initial salt concentration (c_{10} and c_{20}) for both case studies, the salt concentration in the upper unit decreased while irrigation occurred (dilution effect) and increased when irrigation stopped (t = 562 min) (Fig. 4b). Salinity in the lower unit increased as long as water was being depleted by root uptake and did not flow in (water content in the upper unit had not reached FC). A higher rate of salinity decrease during irrigation was clearly associated with a higher initial salt concentration (Fig. 4e). Moreover, for both case studies, the pattern of salinity change in the lower unit differed from that in the upper one. While the decrease in the upper unit was monotonic as long as water flowed in, it varied in a nonmonotonic pattern in the lower unit, first increasing and then decreasing (Fig. 4b, e). A nonmonotonic variation in $\rho(t)$ also occurred in both units for the two case studies (Fig. 4c, f). This nonmonotonicity was a result of the

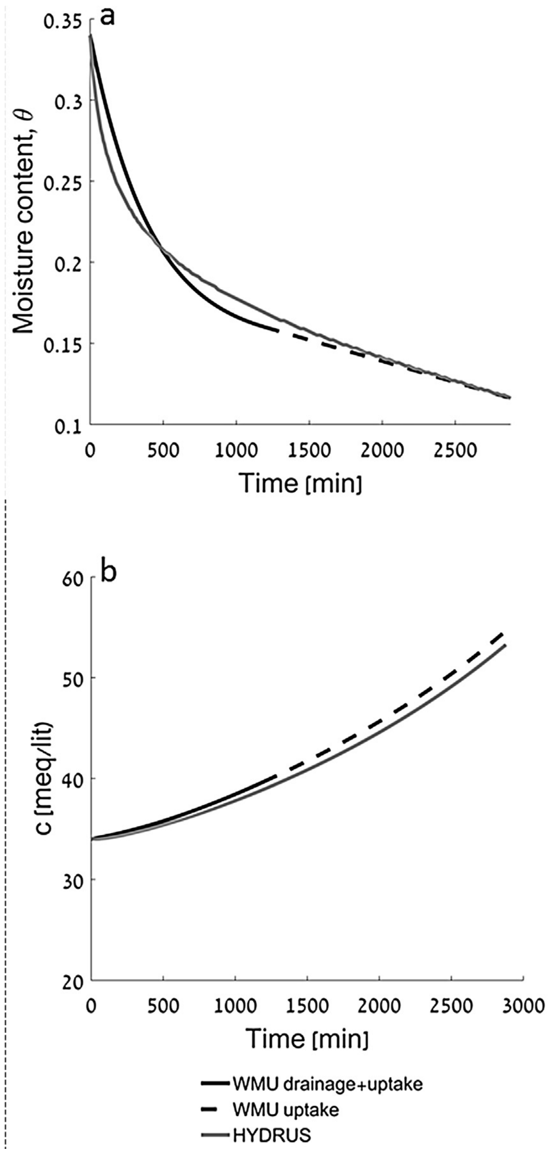

Fig. 3. Comparison of WMU model and HYDRUS 1-D solution for flow and transport averaged over a depth equal to the unit depth. Temporal variation of moisture content (a, c) and salt concentration (b, d) is shown for drainage (a, b) and subsequent irrigation (c, d).

Table 1
Initial conditions and parameters of WMU simulations. Subscripts 1 and 2 for initial moisture content and concentration refer to, respectively, the top and bottom units.

# Simulation	Initial conditions			
	$\theta_{i\,1}$	$c_{i\,1}$ [meq/l]	$\theta_{i\,2}$	$c_{i\,2}$ [meq/l]
1	0.1	75	0.1	75
2	0.1	25	0.1	25
3	0.1	75	0.15	50
4	0.1	25	0.15	75
5	0.1	50	0.15	50
6	0.15	50	0.1	50

opposite effects of θ and c on ρ (Eq. (5)): a decreased salt concentration decreases ER whereas increased moisture content increases it. So although during irrigation, the moisture content rose continuously, the ER(t) increased after an initial decrease, owing to the salt dilution in the unit. This pattern may confuse the interpretation of ER(t) when the changes in solute concentration are not taken into account, which may lead to misinterpretation of moisture content distribution and in particular, oversaturation.

The simulation results for three additional cases are depicted in Fig. 5, where the upper unit was initially drier than the lower one ($\theta_{10} = 0.097$; $\theta_{20} = 0.15$) for all case studies. This represents a common scenario in irrigated fields where the topsoil layer is drier than the layer underneath it before irrigation is started owing to evaporation and enhanced root water uptake. The initial salt concentration in the upper unit was higher than, lower than, or equal to the concentration in the lower unit (Fig. 5b, e and h, respectively). The general pattern of $\theta(t)$ and $c(t)$ variation in Fig. 5 was similar to those in the previous two case studies (Fig. 4a, b, d, e), despite the different initial water content. As the figure speaks for itself, we skip the detailed verbal description of $\theta(t)$ and $c(t)$ variation for the latter case studies. The variation in $\rho(t)$ in Fig. 5c, f, i indicates that the water content variation had a larger effect than that of the salt concentration. All simulations depicted in Fig. 5 showed a similar trend: when water enters the upper unit, $\rho(t)$ first decreases sharply and then shows a gradual or no increase, depending on the balance between the rates of water content increase and salt concentration decrease.

Consequently, the simulations indicated that an a priori estimation of $\rho(t)$ from variations in $\theta(t)$ and $c(t)$ is not straightforward, owing to

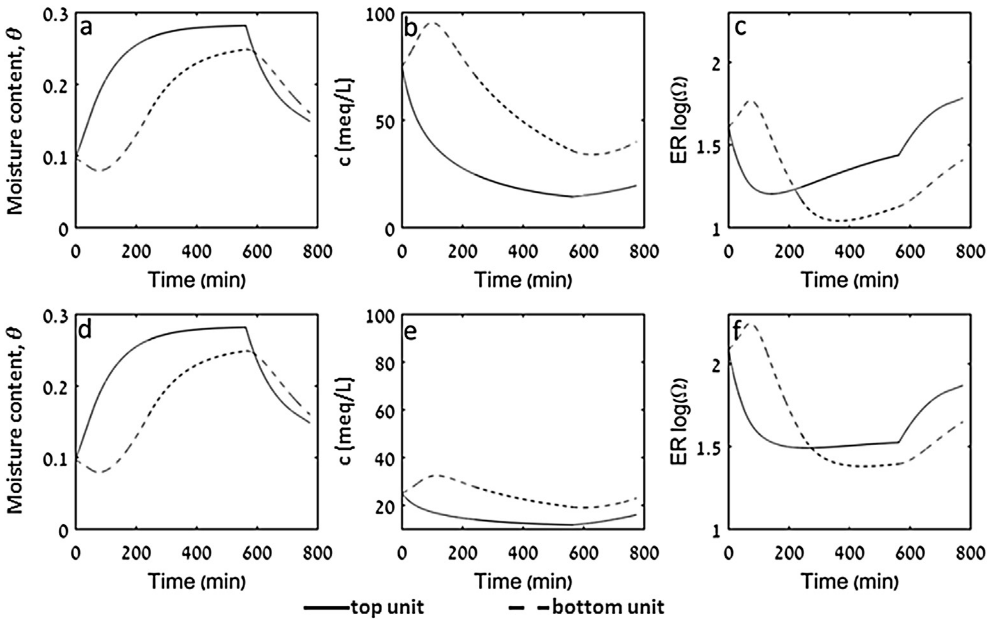

Fig. 4. Variation of moisture content, θ(t), salt concentration, $c(t)$, and electrical resistivity, ER(t), in the upper and lower units of a two-unit system. (a–c) High initial salt concentration simulation. (d–f) Low initial salt concentration simulation. Irrigation with brackish water started at t = 0, while water content in both units was below field capacity (FC). At t = 51 min, brackish water started to drain from the upper unit into the lower one as the water level in the former reached FC. At t = 242 min, water level in the lower unit also reached FC and drainage of brackish water occurred simultaneously from both units. Note that water is continuously withdrawn from both units by plant water uptake.

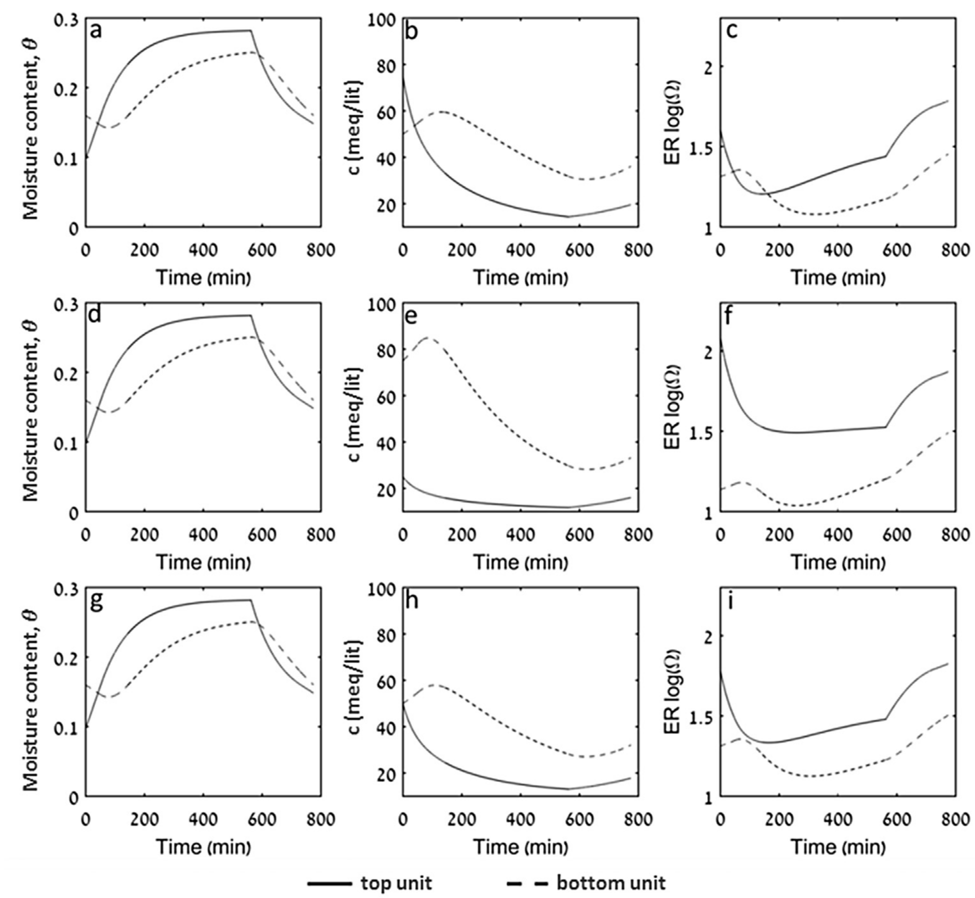

Fig. 5. Variation of moisture content, θ(t), salt concentration, $c(t)$, and electrical resistivity, ER(t), in the upper and lower units of a two-unit system. (a–c) Initially higher salt concentration and lower moisture content in the upper unit vs. lower unit. (d–f) Lower initial moisture content and salt concentration in the upper unit vs. lower unit, (g-i) same initial salt concentration in both units.

the opposite effects of the two latter variables on the former.

3.4. Do preferential ER and preferential flow pathways coincide?

The examination of whether the in situ-observed preferential ER pathways are associated with preferential flow pathways or a result of spatial salinity distribution was based on the aforementioned WMU model simulations (Figs. 4 and 5).

The overall $\mathrm{ER}(t)$ variation within the segments was summarized, based on the simulations (Figs. 4 and 5), for further use in analyzing the field-measured ERT scans, as follows:

i. A monotonic ER(t) decrease was associated with a continuous increase in net water content in the segment. The rate of the ER(t) decrease depended mainly on the initial water content in the segment and the rate at which it increased, and to a lesser extent on the initial salt concentration and its variation over time within the segment.
ii. A monotonic increase of ER(t) was associated with net continuous water content depletion (by ET, drainage, or both). As was demonstrated in case 1, the rate at which the ER(t) increased depended mainly on the rate at which the water was depleted. Salinity variation in the segment had a minor effect on the rate at which the ER(t) increased.
iii. A constant ER(t) indicated no change in water content or salinity.
iv. An ER(t) decrease followed by an increase (nonmonotonic variation) occurred when water entered a segment, while drainage started sometime afterward, namely, when the water level reached FC. The lowest ER(t) value coincided with drainage initiation. As the drainage rate depended on water content in the segment, the rate of its increase diminished as water content increased (Figs. 4a, d, and 5a, d, g), and the ER(t) increase followed the inverse pattern. However, while the moisture content slowly reached a constant value, the salt in the segment underwent a continuous dilution due to a lower input concentration entering from above. This combination caused an increase in ER(t). The increase intensified when the irrigation stopped, and both drainage and ET were occurring. The initial salt concentration in the segment affected the rate at which ER(t) increased, where a higher/lower rate of increase was associated with higher/lower initial salt concentration, respectively. Note that salinity affected the rate at which ER(t) varied, but not the overall variation pattern.
v. An ER(t) increase followed by a decrease (nonmonotonic variation) represented the case in which water is first depleted by ET and is then increased by the recharging water (irrigation into and drainage from an upper segment to a lower one). The ER(t) increase rate depended on the salt concentration in the segment, being higher for higher salt concentration. The decrease in ER(t) initiated when drainage from the upper segment started to flow into the lower one. The rate of ER(t) increase depended on the initial moisture content, the water-depletion rate, the balance between the water discharge and depletion rates, and to a minor extent, on the salt concentration.

The above ER(t) variation scenarios within a certain soil volume show that this variation is mainly dependent on the change in water content and to a lesser extent on the variation in salt concentration.

To determine whether the highlighted columns in Fig. 2. are preferential paths or not, the ER variation with time within each of the three segments in the eight columns, $\mathrm{ER}_{i,j}(t)$, (where $i = 1 - 3$, and $j = 1 - 8$) (Fig. 2), depicted in Fig. 6, was studied. The ER variation in each unit was then compared to the five scenarios above. The variation of ER(t) for each segment was obtained by following the change in the measured ER for the segment in consecutive scans (Fig. 2). The ER for the segment was determined by averaging the ER(x, z) within each segment, with t = 0 in Fig. 6 denotes the start of irrigation. According to Fig. 6, there are nonmonotonic ER(t) patterns in the upper segment (blue line) of sections 1, 2, 3, 4, 5, 7 and 8. The ER(t) in the upper segment of sections 1, 3, 4, 5, 7 and 8 first decreased and then increased (scenario iv above), whereas in section 2, it first increased and then decreased (scenario v above). The ER(t) remained mostly constant in section 6 (scenario iii above). The nonmonotonic ER variation in the lowest units of sections 1, 4, 5 and 7 can be related to the retarded water entrance from the above unit (scenario iv above) and of sections 2, 3, 6 and 8 to a moisture content increase followed by a dilution (scenario v above) (Fig. 6). Notice that the initial retardation in section 7 is also related to scenario v.

Following the above patterns, a distinction between preferential and uniform flow could be made by examining the following sequence of events: (i) the three units in a section receive water in sequential order similar to the gradual wetting front advance, which indicates that flow is stable; (ii) two or three units in a section wet simultaneously, indicating that flow is preferential; (iii) the lower unit receives water before the upper one, indicating a preferential flow path that bypasses the upper unit (the case of a 3D flow pattern that cannot be monitored by the 2D ERT scans). The application of these characteristic patterns to the measured ER(t) scans revealed that the flow in sections 2, 3 and 5 in Fig. 6 is uniform, whereas the flow in sections 1, 4, 6, 7 and 8 is preferential, bypassing the upper unit in sections 1, 7 and 8, or the middle unit in sections 4 and 6.

4. Discussion

4.1. General

Given that salinity and water content affect soil ER independently, albeit not in a similar fashion, the identification of spatial water distribution in the soil profile using ERT becomes a challenging task when brackish water is used. This challenge is even greater when flow takes place in preferential flow paths (fingers) that are inherent to water-repellent soils that are irrigated with TWW that contain high salt concentrations. The current study presents a method that combines ERT scans and a relatively simple mathematical model to identify preferential flow paths in the soil profile for the above-mentioned case. The method is based on frequent ERT scans during an irrigation event and the identification of preferential ER paths. The ER(t) for each of couple vertical soil segments that comprise the preferential ER path are then compared qualitatively to the ER(t) patterns that were independently calculated by the WMU model. The changes within the soil segments are simulated by a series of interconnected well-mixed units that are located one above the other. Given that the spatial salinity concentration within each unit is assumed uniform and varies with time, the spatial variation along the preferential ER path is taken care of by the variation within the interconnected units. The combination of measured and simulated ER(t) scans enabled to examine whether the ER preferential paths were generated by preferential flow or rather by a certain salt distribution. In the current case, using this method verified that the preferential ER(t) pathways coincided with preferential flow pathways.

Additionally, and as opposed to interpreting the ER changes by solving the Richards equation for water and convective-dispersive solvers (e.g. HYDRUS 1-D) along the entire preferential path, a discretization of the preferential ER path into well-mixed units enables to detect flow that bypasses some units above it. Such scenario takes often place under unstable flow conditions. This flow option is obtained when the variation of water content and salt concentration with time in a unit does not obey the variation as calculated by the WMU model. For such case, a 2-D flow is indirectly identified by using the 1-D WMU model.

4.2. Method limitations and guidelines for use

Besides its advantages, the method suggested in this study has some

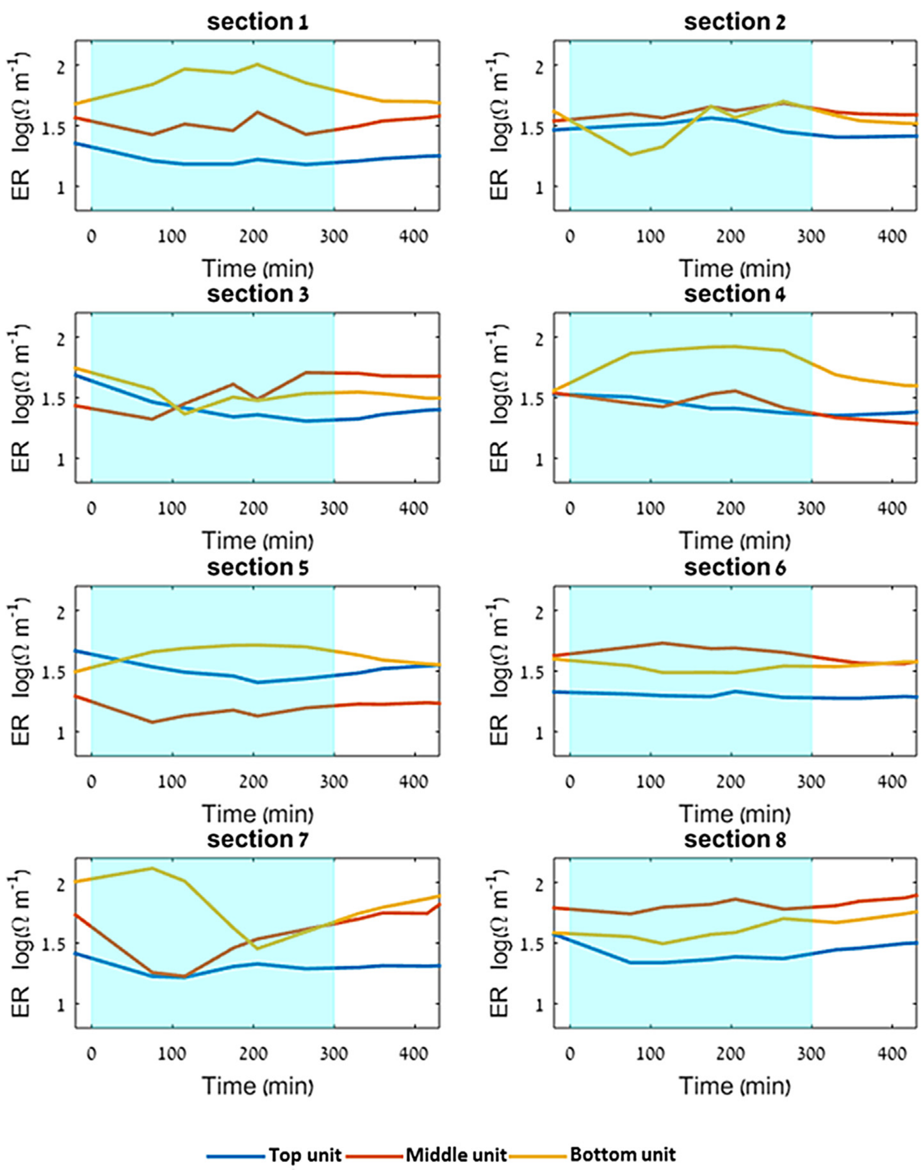

Fig. 6. Temporal changes in electrical resistivity (ER) for three segments in sections isolated from the ERT spatial distribution results (Fig. 2). In each section, the temporal changes in mean ER for the topmost, middle and bottom-most units are presents in blue, red and yellow, respectively. The light teal and white backgrounds represent, respectively, when irrigation was applied and when it was not.

limitations that should be considered prior to its application and results analysis.

The resolution, sensitivity, depth of investigation, and the ability to detect particular ER distribution depend on the purpose of the survey as they are affected by the electrode configuration, spacing, and length (Binley and Kemna, 2005; Samouëlian et al., 2005). For example, the dipole-dipole configuration has high sensitivity to changes in the horizontal direction, whereas the Wenner–Schlumberger configuration has a higher sensitivity to changes in the vertical direction. However, for all configurations, the resolution and sensitivity of the ER survey decrease as the spacing between electrode increases. A practicable approach to deal with this issue is to conduct a series of forward problem simulations using different ERT surveys of different spacing and configurations (Furman et al., 2007; Rings and Hauck, 2009; Garré et al., 2012). A theoretical study that investigated adequate electrode configurations for preferential flow pathways detection in soils (Brindt, 2012) revealed that for the given electrode spacing and survey configuration (dipole-dipole) fingers of 5–10 cm width and higher can be identified. However, an ERT survey that uses the above-mentioned configurations with all possible electrode quadripoles will yield several thousand measurements for only a few electrode quadripoles, and will substantially increase the acquisition time (Garré et al., 2012), and will not be practical for repeatable ER scans during soil wetting and drainage.

An inherent drawback of the WMU model is that only flow in the vertical direction can be simulated as the input to, and output from each unit. This assumption limits the use of the WMU model for 1D that takes place within a preferential ER path rather than for 2D. Moreover, the accuracy of the WMU model that assumes well-mixed units depends on the unit depth, b. The accuracy increases as more units are used. The number of units should be chosen by a compromise between accuracy and practical feasibilities. Units of 20 cm depth each were chosen as the residence time in a unit that simulates sandy soil is relatively short, as gravity driving force is larger than capillary driving force. Thus, a higher number of units should be considered for soils of lower permeability. The values chosen for α and FC parameters should also be related to the associated hydraulic properties of the simulated soil.

Regarding the relationship between the pore water electrical conductivity and salt concentration in the Archie law, it should be noted the linear relationship that used in the current study is valid only for low concentrations (pore water electrical conductivity up to $10 dSm^{-1}$). A non-linear relationship should be considered for higher salt concentrations (Shea and Luthin, 1961). Solute composition may affect the electrical properties as solute charge and mobility are playing a key role (Samouëlian et al., 2005).

5. Conclusions and recommendations

This work supports the use of the ERT method for locating and tracking of gravity-driven fingering in water repellent plots that are irrigated with TWW. Soil water salinity may cause difficulties in finding fingers owing to its influence on soil electrical resistivity. However, analyzing the ER(t) results by using WMU model simulations enables the isolation of gravity-driven fingers with higher confidence.

The suggested method can be used for studying the water distribution in the soil profile, and particularly if preferential ER has been measured. The knowledge whether measured preferential ER paths are due to preferential flow or rather to a certain spatial salt distribution is important if treatments to ease their effects are sought. In both cases, an appropriate irrigation design may help. In cases where preferential ER paths coincide with preferential flow pathways, a particular irrigation design and management can ease the negative effects of such flow regime on soil water availability to plant roots and the enhanced leaching of agrochemical below the active root zone. A customized, although different, irrigation design can ease the negative effects associated with salinity-induced preferential ER paths by leaching the salt below the root zone. Beyond identification of the cause to preferential ER paths, repeated ER measurements enable to monitor the effectiveness of the methods used to reclaim the current situation, and improve it if found not efficient.

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